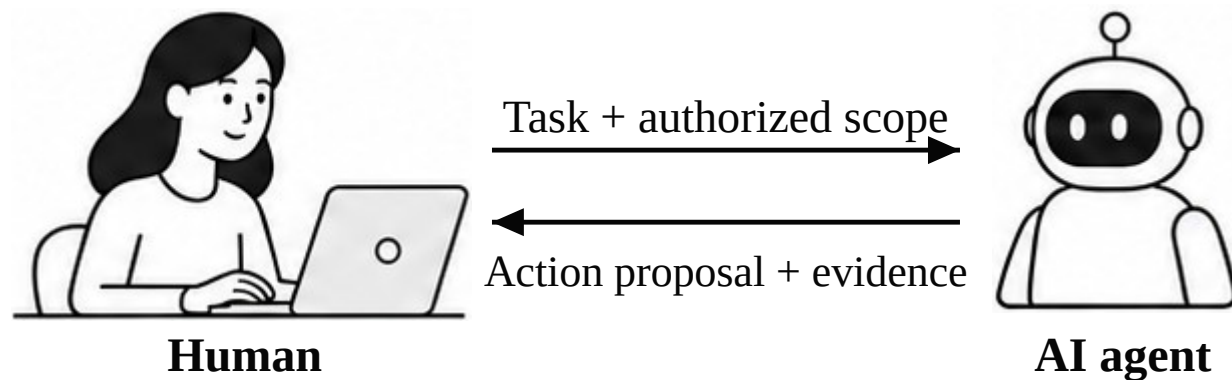


Reliable AI Agents Under Human Control

A proposed research agenda for actions, system updates, and delegation

Human–Agent Interaction

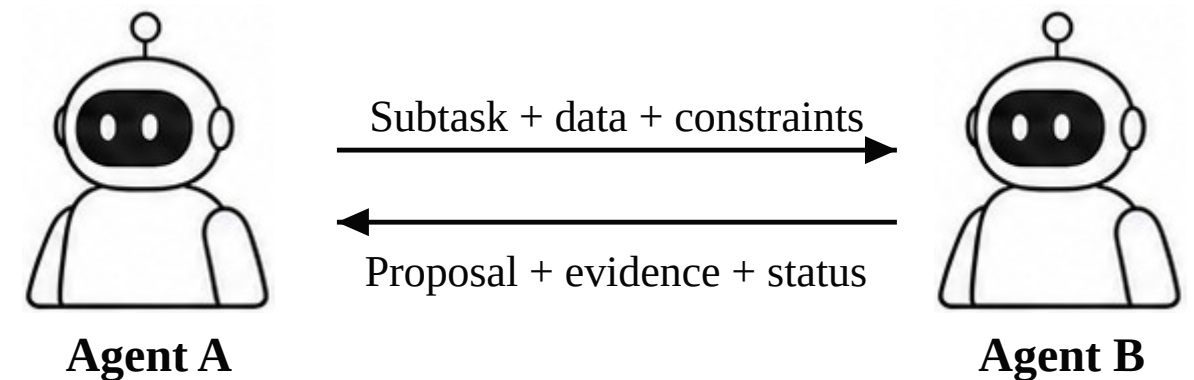


I Check proposed actions against approved requirements

II Evaluate security and reliability after system updates

III Enforce current scope and permissions across task steps

Agent–Agent Collaboration



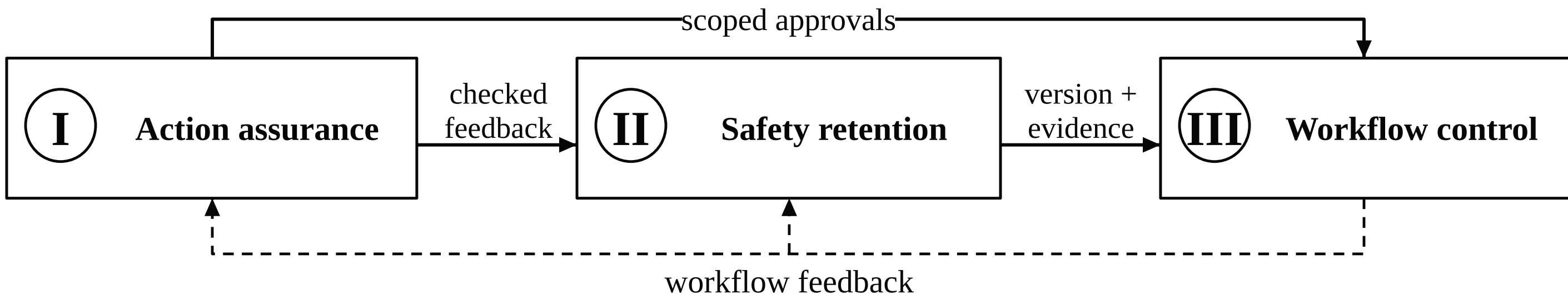
I Check proposed actions using independent evidence

II Assess retention after either agent or harness updates

III Enforce inherited data limits and valid delegation scope

Shared human-approved requirements Φ

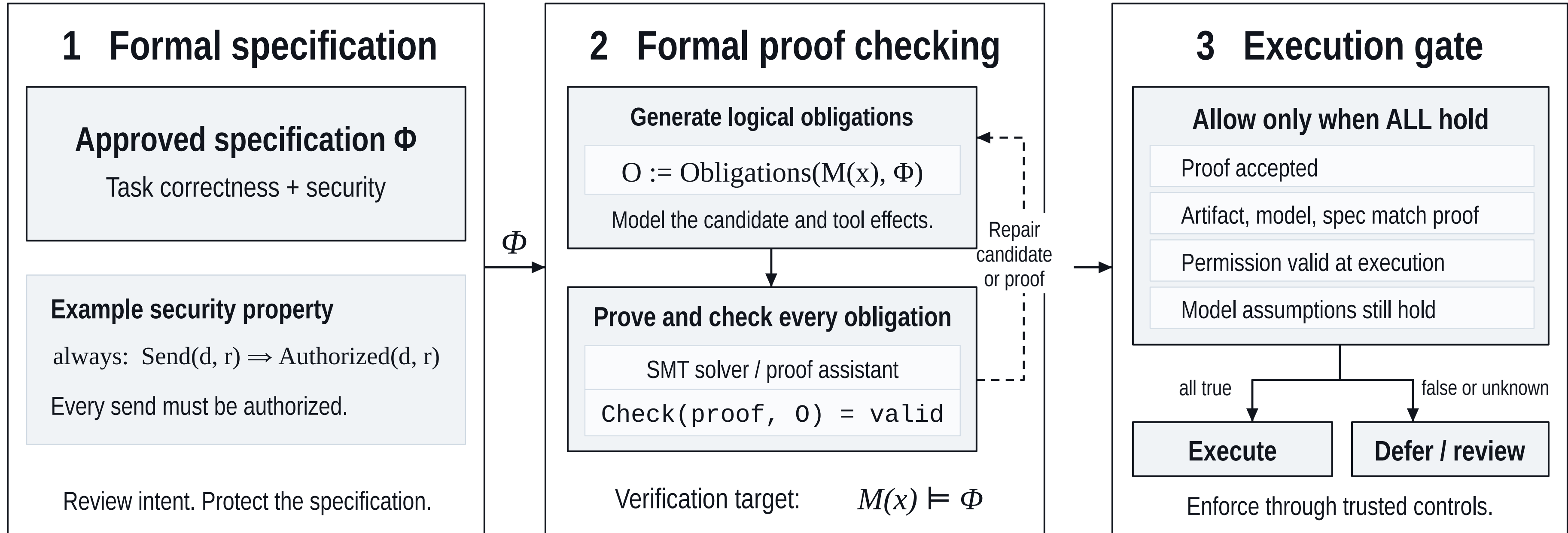
Task criteria · security properties · authorization scope



Formal claims require faithful models, sound checking, and trusted enforcement;
tests provide evidence for evaluated conditions.

Formal Oversight Before Action

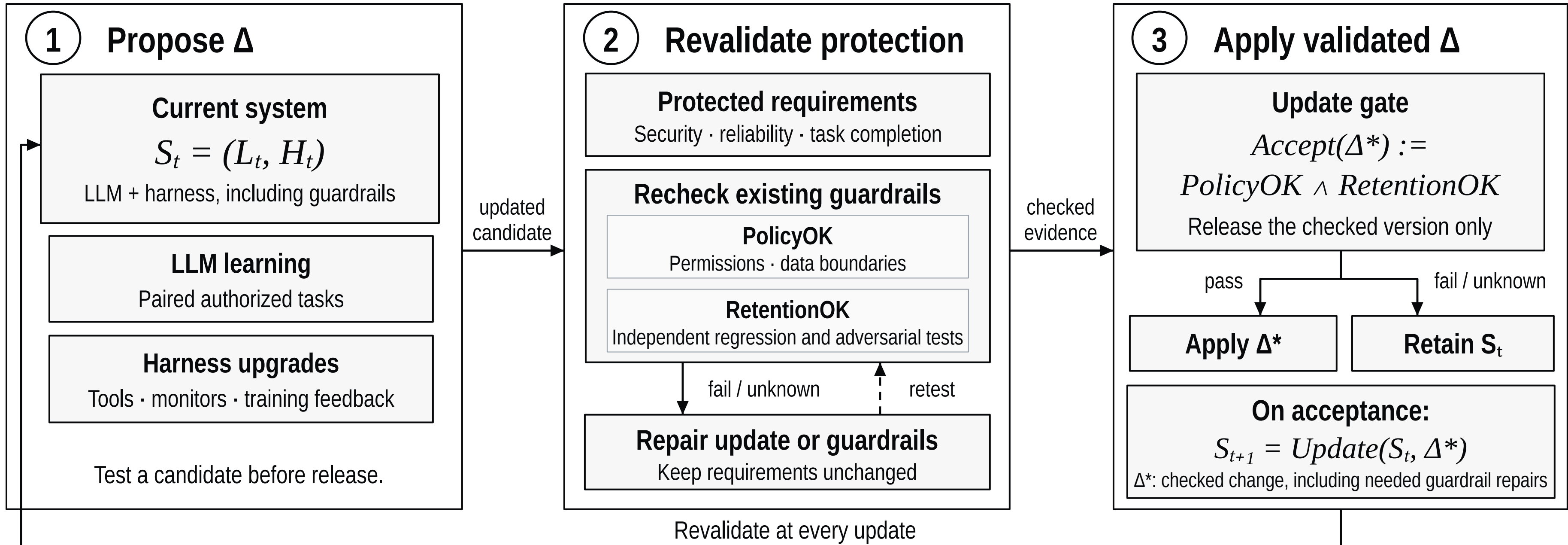
Proposed framework: specify properties, prove obligations, enforce execution



Assurance requires faithful models, sound proof checking, and trusted enforcement.

Preserving Safety Through Continual Updates

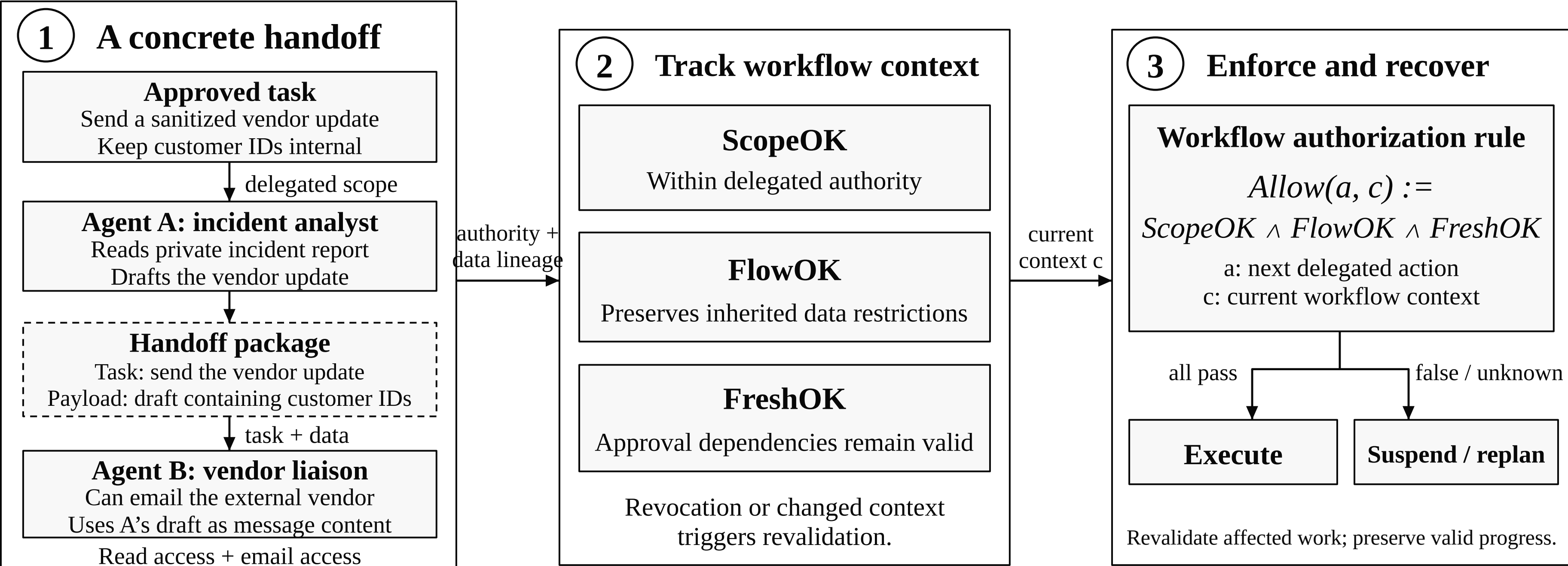
LLM or harness changes can invalidate existing guardrails



Protection requirements stay fixed; guardrail implementations may evolve.

III. Authorization Across Delegated Workflows

A private report can leak through a delegated email draft.



Guarantees depend on captured effects and trusted enforcement.